



TEIAS | Tehran Institute for
Advanced Studies

Econometrics II

HW 1: Let's dive deeper into R

Due: Friday, Esfand 25, 23:59

Instructions

- Please name your file `lastname.firstname.HW#.zip`.
- Email your work to airanpour22@gmail.com and unesrom@gmail.com before the deadline.
- In this homework, you need the datasets of the Labor Force Survey (LFS), which you can download from the website of the Statistical Center of Iran.
- You should definitely analyze the data related to this exercise with R and send me your codes in R markdown and report in html format.
- All requested tables and visualizations should be clear, appropriately labeled, and include legends when necessary. Be prepared to discuss the implications of your findings.
- Pay attention to the reproducibility of your code

Project Tasks

Your project will involve extensive data analysis using the Labor Force Survey (LFS) dataset [for the years 1397-1401](#). The objectives are to understand workforce participation trends, sectoral employment, and gender dynamics in the labor market. Below are the tasks you are expected to complete:

1. Workforce Participation Analysis. (15 points)

- Identify the three provinces with the highest and lowest participation rates in the year 1397.
- Compare these rates with the national average for the same year and report the results.
- Visualize the participation rates for these provinces and the whole country from 1397 to 1401.

2. Gender-Specific Participation Rates. (25 points)

- Calculate the male and female participation rates in the provinces with the highest and lowest rates, and compare them with the national rates for the year 1397 and report the results.
- Plot the participation rates for both genders through these years, illustrating any trends or disparities.

3. Sectoral Employment and Work Hours. (30 points)

- Determine the female and male share in the three sectors asked in LFS.
- Find the average hours worked by each gender in these sectors.
- Plot the average hours worked alongside the gender share for each sector from 1397 to 1401.

4. Reasons for Job Leave. (20 points)

- Analyze the dataset to extract the top reasons for job leave among males and females in the year 1400 and plot the results.
- Analyze the dataset to extract the top reasons for job leave among single and married females in the year 1400 and plot the results.

5. Gender Gap Analysis. (10 point)

- Discuss potential reasons for the gender gap in workforce participation between male and female workers.